Memoria Justificativa y Documentación Técnica

Ejercicio 8 - Diseño e Implementación de Data Warehouse (Star Schema)

1. Estructura lógica y modelo en estrella (Star Schema)

En este proyecto se ha diseñado un almacén de datos (Data Warehouse) basado en un esquema en estrella (Star Schema) para el análisis de transacciones financieras. El modelo consta de una tabla central de hechos (fact_transactions) y cuatro tablas de dimensiones principales:

• dim_user: Contiene la información unificada de los usuarios.
• dim_company: Información de las empresas y comercios.
• dim_credit_card: Detalles de las tarjetas de crédito y su relación con los usuarios.
• dim_date: Calendario analítico derivado de las marcas de tiempo de las transacciones.

2. Integración de usuarios y definición de la columna region

Para la creación de la dimensión dim_user, se han combinado los datos de los ficheros american_users.csv y european_users.csv. Con el fin de mantener la trazabilidad geográfica de los usuarios, se ha añadido la columna region, asignando el valor 'US' a los usuarios americanos y 'EU' a los europeos.

3. Relaciones e integridad referencial (Foreign Keys)

Para garantizar la integridad referencial de la base de datos:

• La clave primaria id de dim_user se define como clave foránea (user_id) en la tabla dim_credit_card (relación 1 a N).
• Las claves primarias de las dimensiones (dim_user, dim_company, dim_credit_card, dim_date) actúan como claves foráneas en la tabla central fact_transactions.

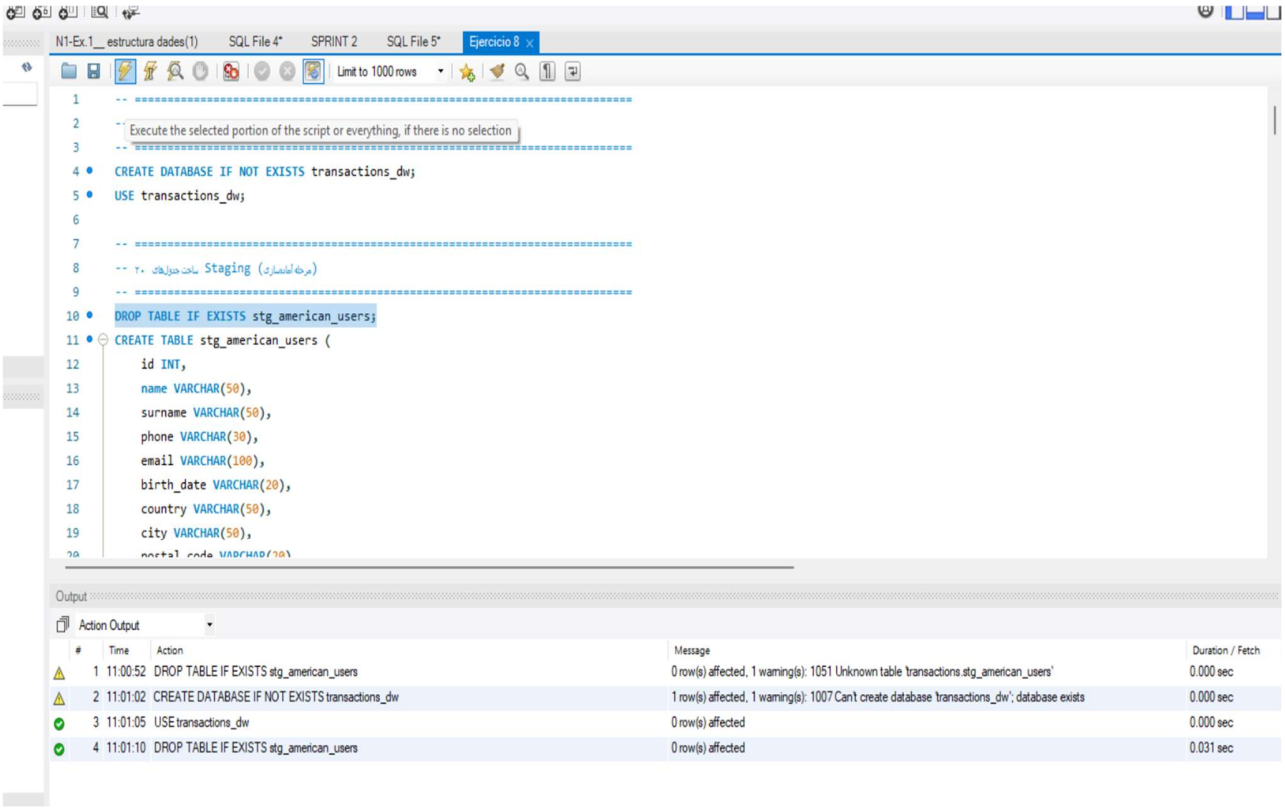
4. Tratamiento del campo de productos (product_ids)

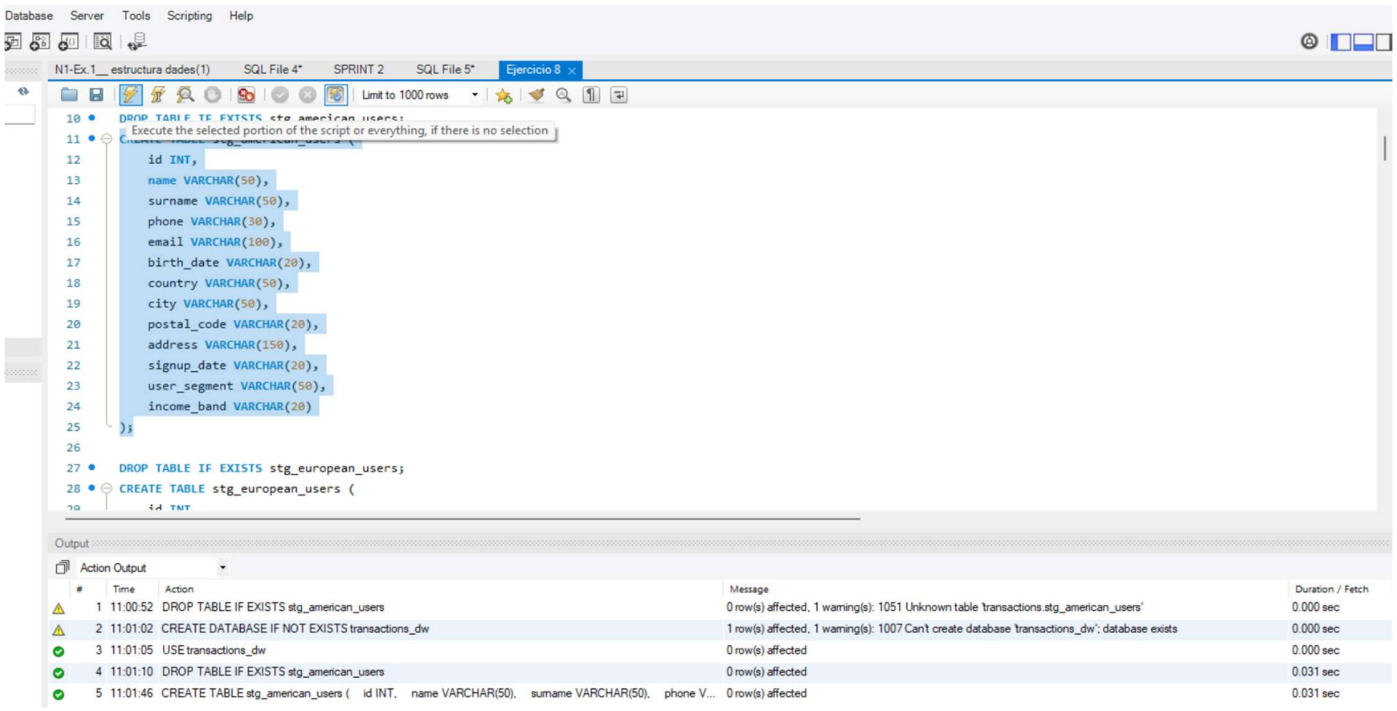
La columna product_ids en la tabla central fact_transactions se ha mantenido en formato de texto (VARCHAR). Esta decisión permite conservar la lista original de identificadores de producto asociados a cada transacción sin alterar el grano de la tabla de hechos, facilitando su posterior procesamiento o desglose en fases avanzadas del proyecto.

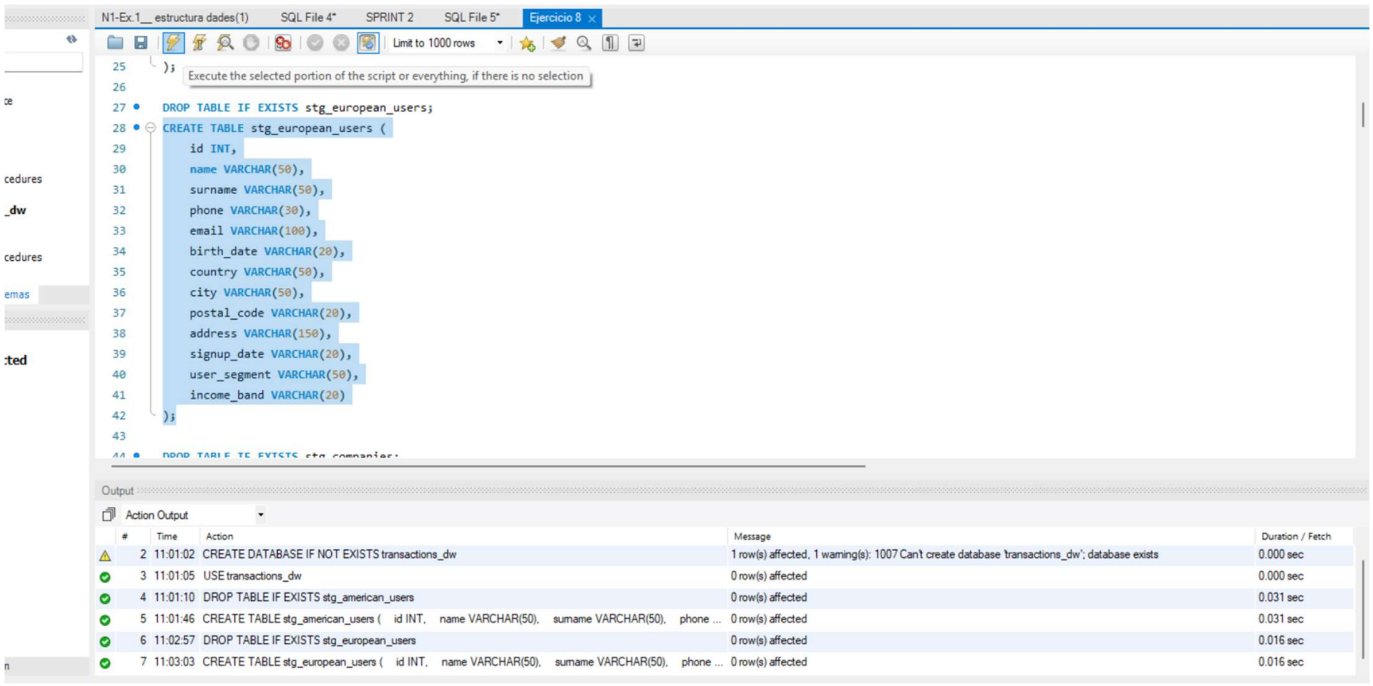
5. Proceso y orden de carga (Pipeline ETL)

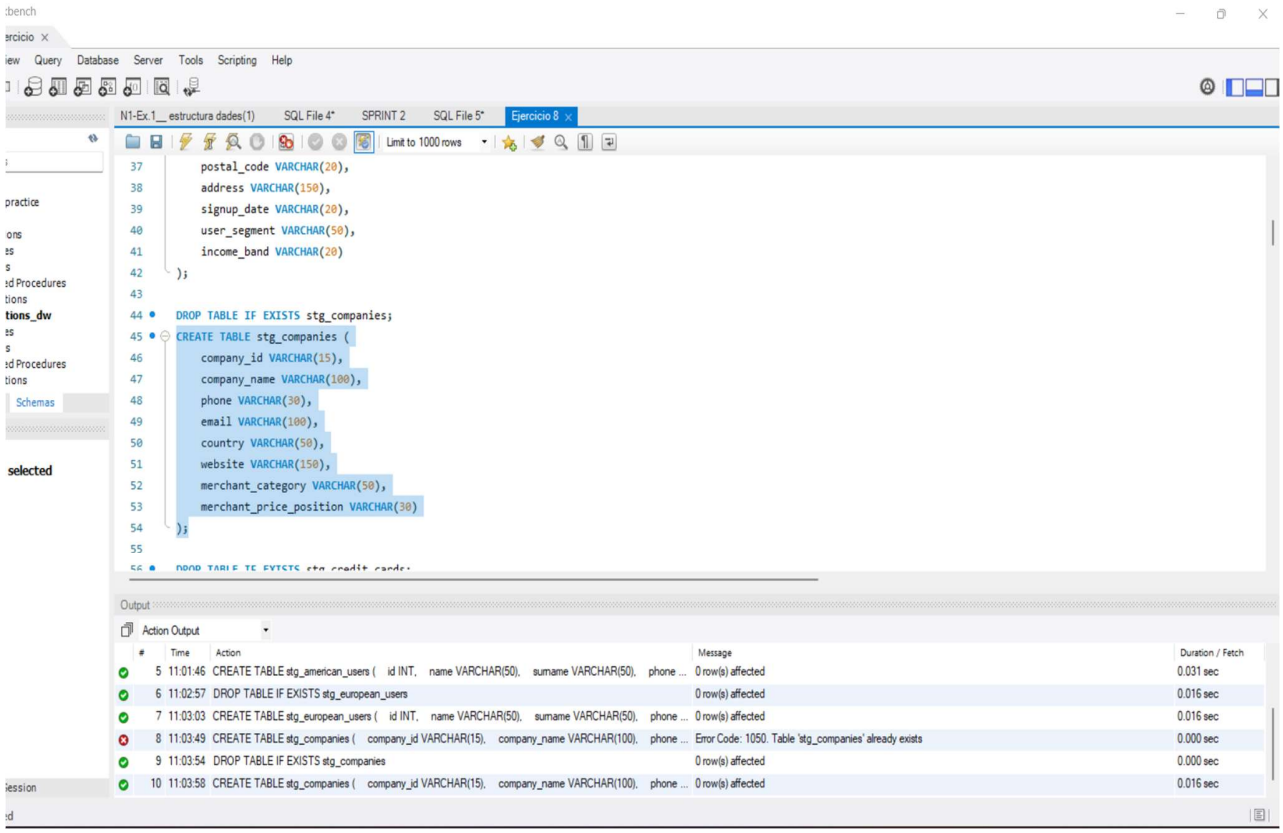
Con el objetivo de evitar conflictos de dependencias por claves foráneas, la ejecución de los scripts sigue un orden estrictamente secuencial:

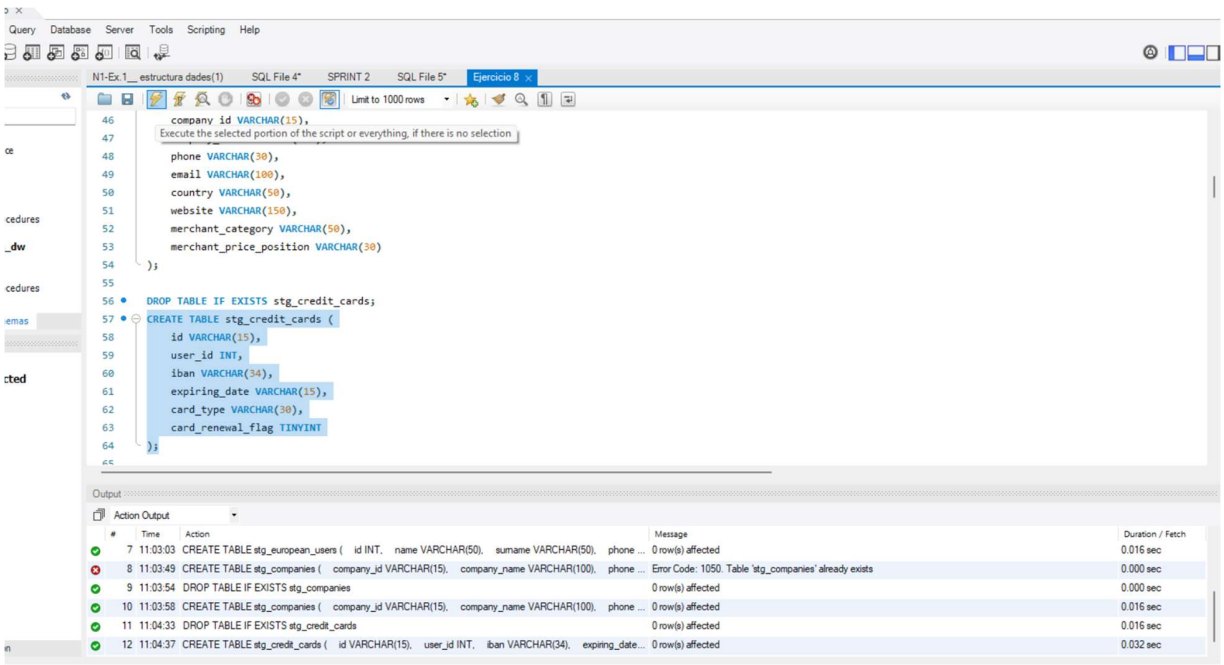
1. Creación de la base de datos transactions_dw y de las tablas de preparación (stg_*).
2. Carga de los datos brutos desde los archivos CSV a la zona de staging.
3. Definición de la estructura DDL de las tablas finales del modelo dimensional.
4. Carga de las dimensiones independientes (dim_user y dim_company).
5. Carga de la dimensión dependiente (dim_credit_card) y del calendario (dim_date).
6. Carga final de la tabla central de hechos (fact_transactions).

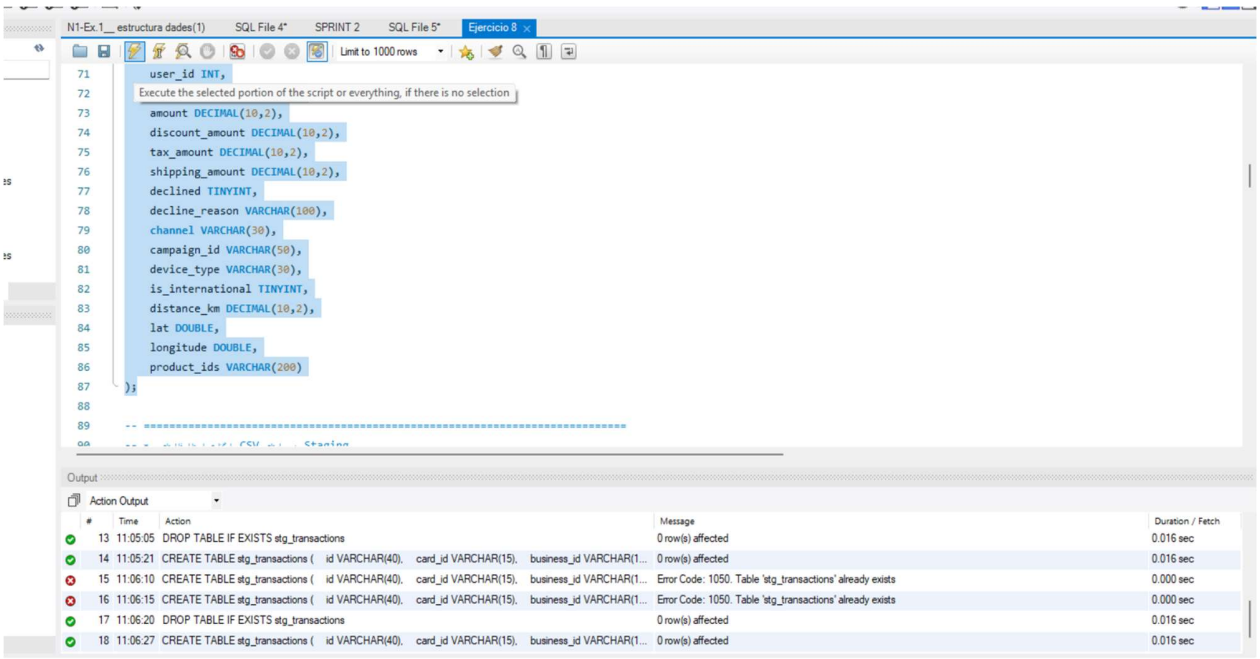


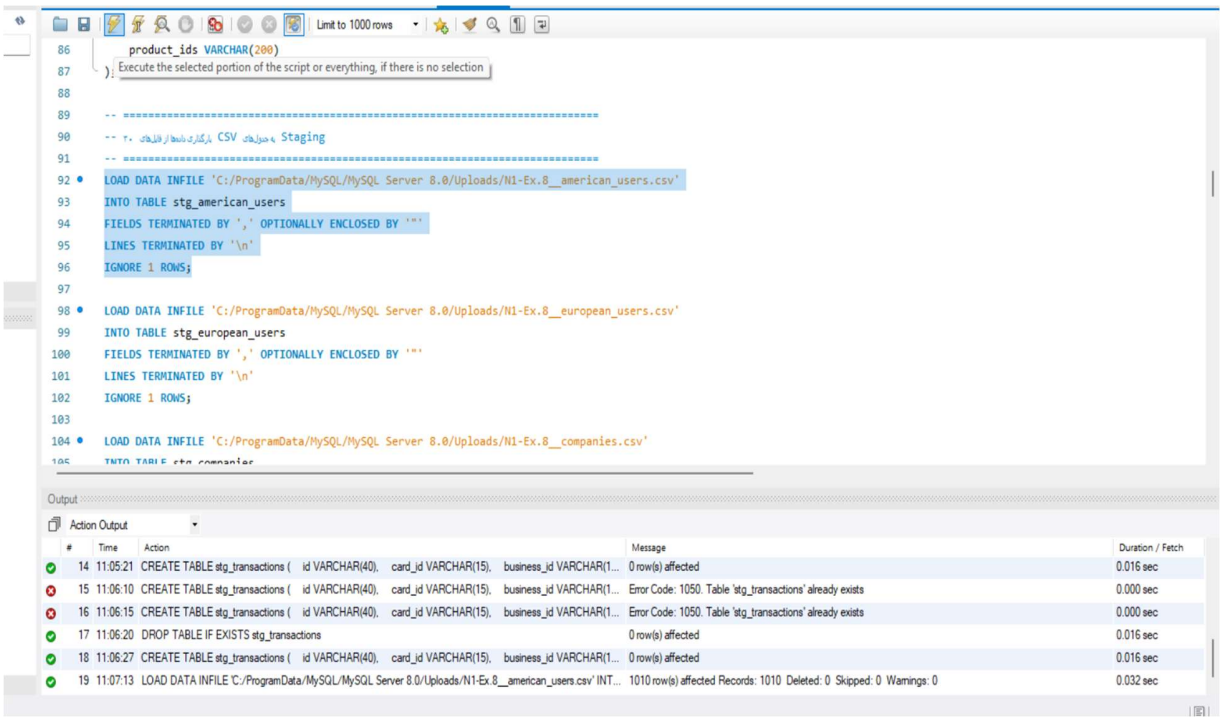

Database Server Tools Scripting Help
N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
Execute the selected portion of the script or everything, if there is no selection
DROP TABLE IF EXISTS stg_american_users;
CREATE TABLE stg_american_users (
id INT,
name VARCHAR(50),
surname VARCHAR(50),
phone VARCHAR(30),
email VARCHAR(100),
birth_date VARCHAR(20),
country VARCHAR(50),
city VARCHAR(50),
postal_code VARCHAR(20),
address VARCHAR(150),
signup_date VARCHAR(20),
user_segment VARCHAR(50),
income_band VARCHAR(20)
);
DROP TABLE IF EXISTS stg_european_users;
CREATE TABLE stg_european_users (
Output
Action Output
Time Action Message Duration / Fetch
1 11:00:52 DROP TABLE IF EXISTS stg_american_users 0 row(s) affected, 1 warning(s): 1051 Unknown table 'transactions.stg_american_users' 0.000 sec
2 11:01:02 CREATE DATABASE IF NOT EXISTS transactions_dw 1 row(s) affected, 1 warning(s): 1007 Can't create database 'transactions_dw'; database exists 0.000 sec
3 11:01:05 USE transactions_dw 0 row(s) affected 0.000 sec
4 11:01:10 DROP TABLE IF EXISTS stg_american_users 0 row(s) affected 0.031 sec
5 11:01:46 CREATE TABLE stg_american_users (id INT, name VARCHAR(50), surname VARCHAR(50), phone V... 0 row(s) affected 0.031 sec

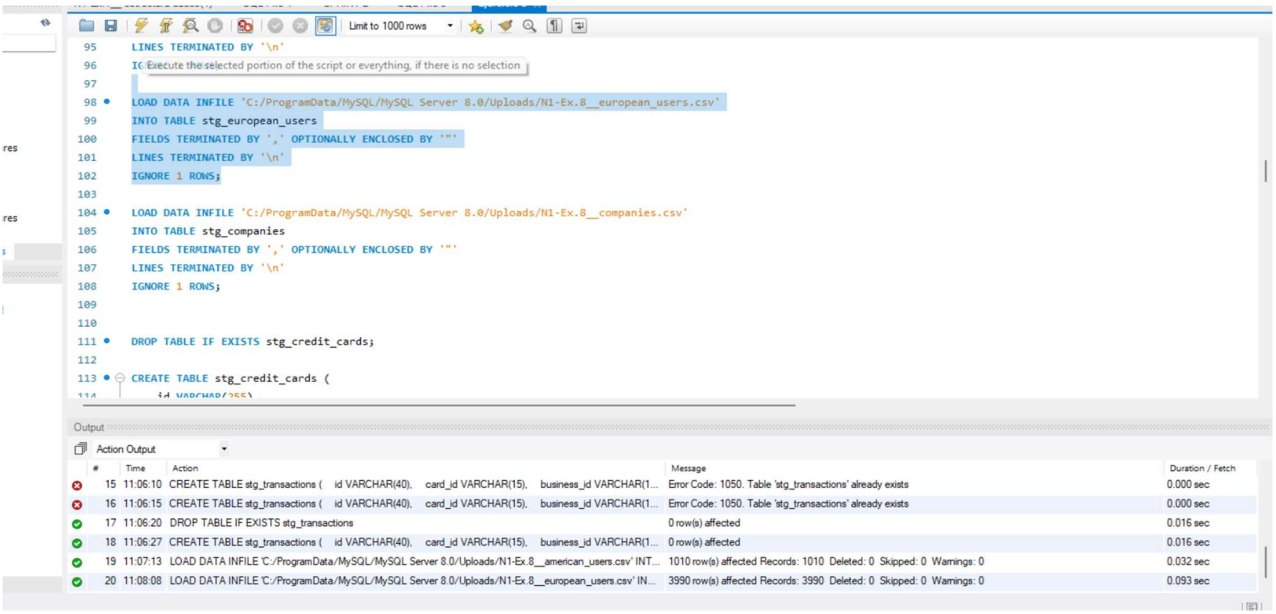

N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
);
Execute the selected portion of the script or everything, if there is no selection
DROP TABLE IF EXISTS stg_european_users;
CREATE TABLE stg_european_users (
id INT,
name VARCHAR(50),
surname VARCHAR(50),
phone VARCHAR(30),
email VARCHAR(100),
birth_date VARCHAR(20),
country VARCHAR(50),
city VARCHAR(50),
postal_code VARCHAR(20),
address VARCHAR(150),
signup_date VARCHAR(20),
user_segment VARCHAR(50),
income_band VARCHAR(20)
);
Output
Action Output
Time Action Message Duration / Fetch
2 11:01:02 CREATE DATABASE IF NOT EXISTS transactions_dw 1 row(s) affected, 1 warning(s): 1007 Can't create database 'transactions_dw'; database exists 0.000 sec
3 11:01:05 USE transactions_dw 0 row(s) affected 0.000 sec
4 11:01:10 DROP TABLE IF EXISTS stg_american_users 0 row(s) affected 0.031 sec
5 11:01:46 CREATE TABLE stg_american_users (id INT, name VARCHAR(50), surname VARCHAR(50), phone ... 0 row(s) affected 0.031 sec
6 11:02:57 DROP TABLE IF EXISTS stg_european_users 0 row(s) affected 0.016 sec
7 11:03:03 CREATE TABLE stg_european_users (id INT, name VARCHAR(50), surname VARCHAR(50), phone ... 0 row(s) affected 0.016 sec


Query Database Server Tools Scripting Help
N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
postal_code VARCHAR(20),
address VARCHAR(150),
signup_date VARCHAR(20),
user_segment VARCHAR(50),
income_band VARCHAR(20)
);
DROP TABLE IF EXISTS stg_companies;
CREATE TABLE stg_companies (
company_id VARCHAR(15),
company_name VARCHAR(100),
phone VARCHAR(30),
email VARCHAR(100),
country VARCHAR(50),
website VARCHAR(150),
merchant_category VARCHAR(50),
merchant_price_position VARCHAR(30)
);
Output
Action Output
Time Action Message Duration / Fetch
5 11:01:46 CREATE TABLE stg_american_users (id INT, name VARCHAR(50), surname VARCHAR(50), phone ... 0 row(s) affected 0.031 sec
6 11:02:57 DROP TABLE IF EXISTS stg_european_users 0 row(s) affected 0.016 sec
7 11:03:03 CREATE TABLE stg_european_users (id INT, name VARCHAR(50), surname VARCHAR(50), phone ... 0 row(s) affected 0.016 sec
8 11:03:49 CREATE TABLE stg_companies (company_id VARCHAR(15), company_name VARCHAR(100), phone ... Error Code: 1050. Table 'stg_companies' already exists 0.000 sec
9 11:03:54 DROP TABLE IF EXISTS stg_companies 0 row(s) affected 0.000 sec
10 11:03:58 CREATE TABLE stg_companies (company_id VARCHAR(15), company_name VARCHAR(100), phone ... 0 row(s) affected 0.016 sec


Query Database Server Tools Scripting Help
N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
company_id VARCHAR(15),
Execute the selected portion of the script or everything, if there is no selection
phone VARCHAR(30),
email VARCHAR(100),
country VARCHAR(50),
website VARCHAR(150),
merchant_category VARCHAR(50),
merchant_price_position VARCHAR(30)
);
DROP TABLE IF EXISTS stg_credit_cards;
CREATE TABLE stg_credit_cards (
id VARCHAR(15),
user_id INT,
iban VARCHAR(34),
expiring_date VARCHAR(15),
card_type VARCHAR(30),
card_renewal_flag TINYINT
);
Output
Action Output
Time Action Message Duration / Fetch
7 11:03:03 CREATE TABLE stg_european_users (id INT, name VARCHAR(50), surname VARCHAR(50), phone ... 0 row(s) affected 0.016 sec
8 11:03:49 CREATE TABLE stg_companies (company_id VARCHAR(15), company_name VARCHAR(100), phone ... Error Code: 1050. Table 'stg_companies' already exists 0.000 sec
9 11:03:54 DROP TABLE IF EXISTS stg_companies 0 row(s) affected 0.000 sec
10 11:03:58 CREATE TABLE stg_companies (company_id VARCHAR(15), company_name VARCHAR(100), phone ... 0 row(s) affected 0.016 sec
11 11:04:33 DROP TABLE IF EXISTS stg_credit_cards 0 row(s) affected 0.016 sec
12 11:04:37 CREATE TABLE stg_credit_cards (id VARCHAR(15), user_id INT, iban VARCHAR(34), expiring_date... 0 row(s) affected 0.032 sec


N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
user_id INT,
Execute the selected portion of the script or everything, if there is no selection
amount DECIMAL(10,2),
discount_amount DECIMAL(10,2),
tax_amount DECIMAL(10,2),
shipping_amount DECIMAL(10,2),
declined TINYINT,
decline_reason VARCHAR(100),
channel VARCHAR(30),
campaign_id VARCHAR(50),
device_type VARCHAR(30),
is_international TINYINT,
distance_km DECIMAL(10,2),
lat DOUBLE,
longitude DOUBLE,
product_ids VARCHAR(200)
);
-- ==========
Output
Action Output
Time Action Message Duration / Fetch
13 11:05:05 DROP TABLE IF EXISTS stg_transactions 0 row(s) affected 0.016 sec
14 11:05:21 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... 0 row(s) affected 0.016 sec
15 11:06:10 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... Error Code: 1050. Table 'stg_transactions' already exists 0.000 sec
16 11:06:15 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... Error Code: 1050. Table 'stg_transactions' already exists 0.000 sec
17 11:06:20 DROP TABLE IF EXISTS stg_transactions 0 row(s) affected 0.016 sec
18 11:06:27 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... 0 row(s) affected 0.016 sec


Limit to 1000 rows
product_ids VARCHAR(200)
);
Execute the selected portion of the script or everything, if there is no selection
-- ==========
-- ۲. بارگذاری داده‌ها از فایل‌های CSV به جدول‌های Staging
-- ==========
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv'
INTO TABLE stg_american_users
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv'
INTO TABLE stg_european_users
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv'
Output
Action Output
Time Action Message Duration / Fetch
14 11:05:21 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... 0 row(s) affected 0.016 sec
15 11:06:10 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... Error Code: 1050. Table 'stg_transactions' already exists 0.000 sec
16 11:06:15 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... Error Code: 1050. Table 'stg_transactions' already exists 0.000 sec
17 11:06:20 DROP TABLE IF EXISTS stg_transactions 0 row(s) affected 0.016 sec
18 11:06:27 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... 0 row(s) affected 0.016 sec
19 11:07:13 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv' INT... 1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0 0.032 sec



```
LINES TERMINATED BY '\n'
IG Execute the selected portion of the script or everything, if there is no selection

LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv'
INTO TABLE stg_european_users
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv'
INTO TABLE stg_companies
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

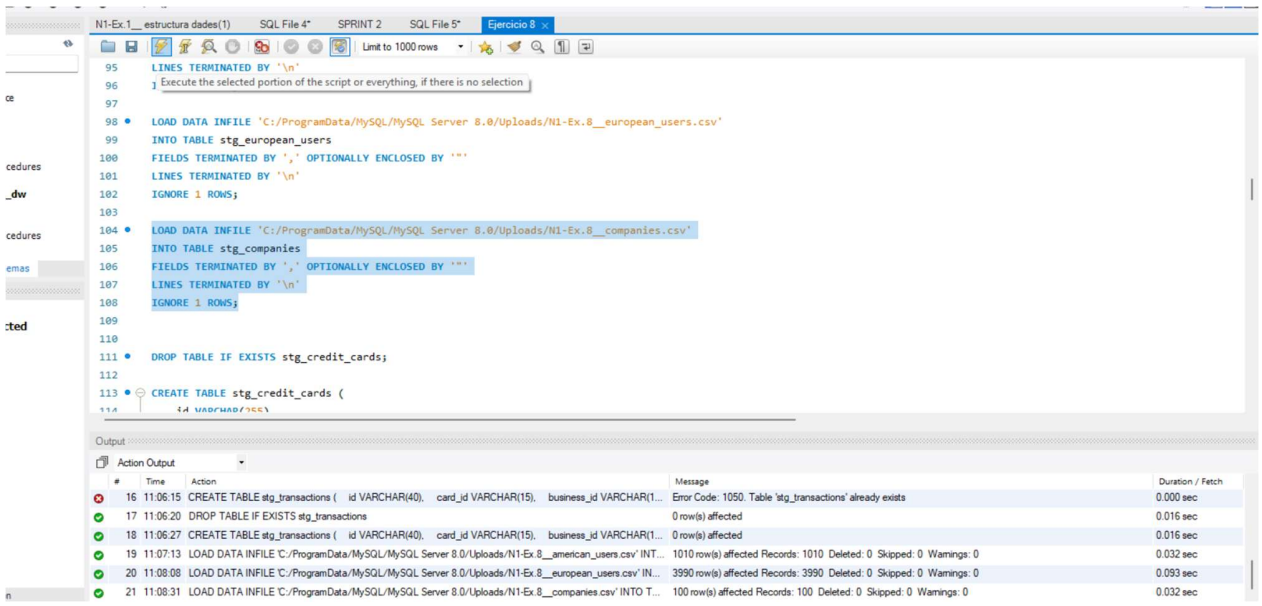

DROP TABLE IF EXISTS stg_credit_cards;

CREATE TABLE stg_credit_cards (
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
15	11:06:10	CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1...	Error Code: 1050. Table 'stg_transactions' already exists	0.000 sec
16	11:06:15	CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1...	Error Code: 1050. Table 'stg_transactions' already exists	0.000 sec
17	11:06:20	DROP TABLE IF EXISTS stg_transactions	0 row(s) affected	0.016 sec
18	11:06:27	CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1...	0 row(s) affected	0.016 sec
19	11:07:13	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv' INT...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0	0.032 sec
20	11:08:08	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv' IN...	3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0	0.093 sec



N1-Ex.1__ estructura dades(1) | SQL File 4* | SPRINT 2 | SQL File 5* | Ejercicio 8

```
LINES TERMINATED BY '\n'
IG Execute the selected portion of the script or everything, if there is no selection

LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv'
INTO TABLE stg_european_users
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv'
INTO TABLE stg_companies
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

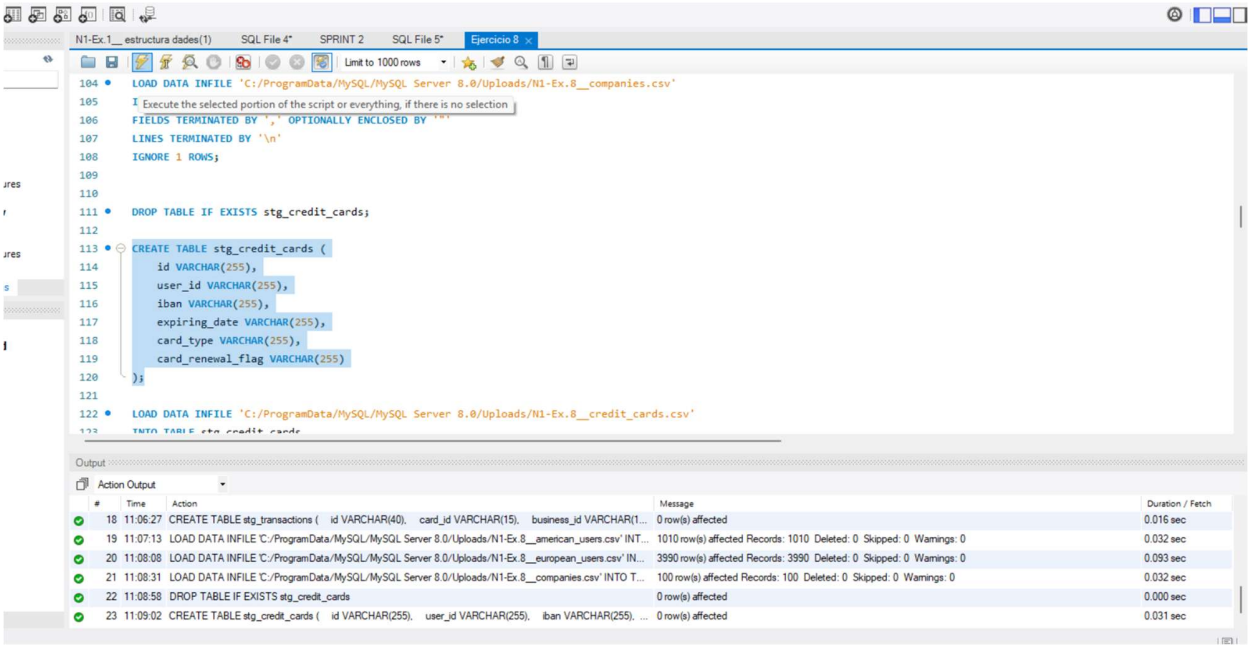

DROP TABLE IF EXISTS stg_credit_cards;

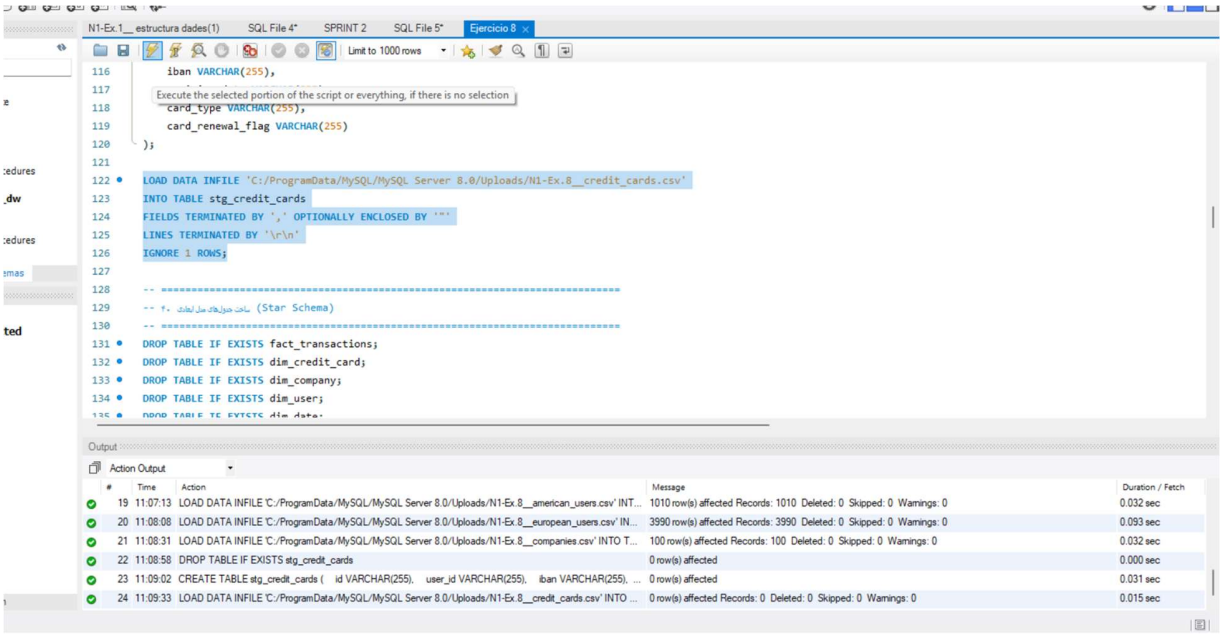
CREATE TABLE stg_credit_cards (
```

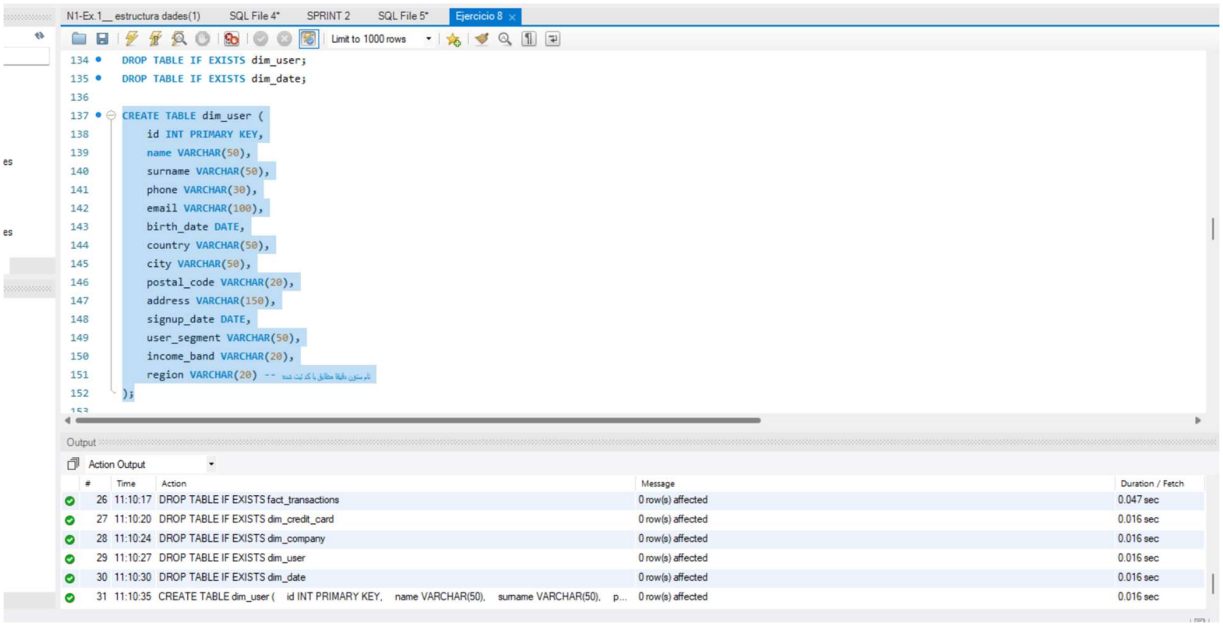
Output

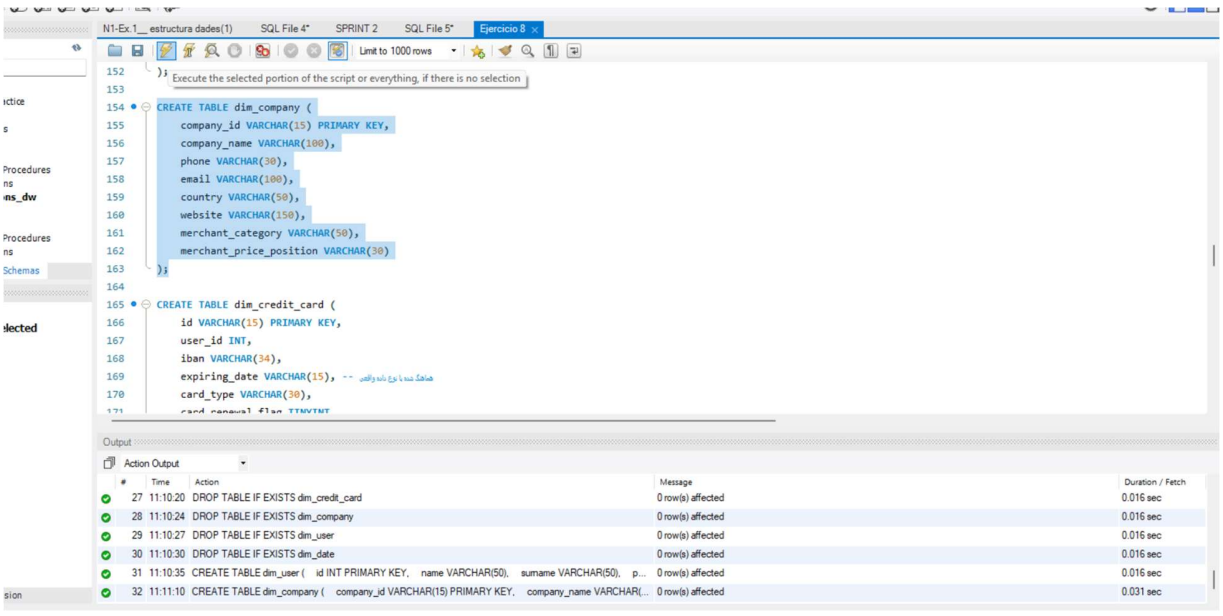
Action Output

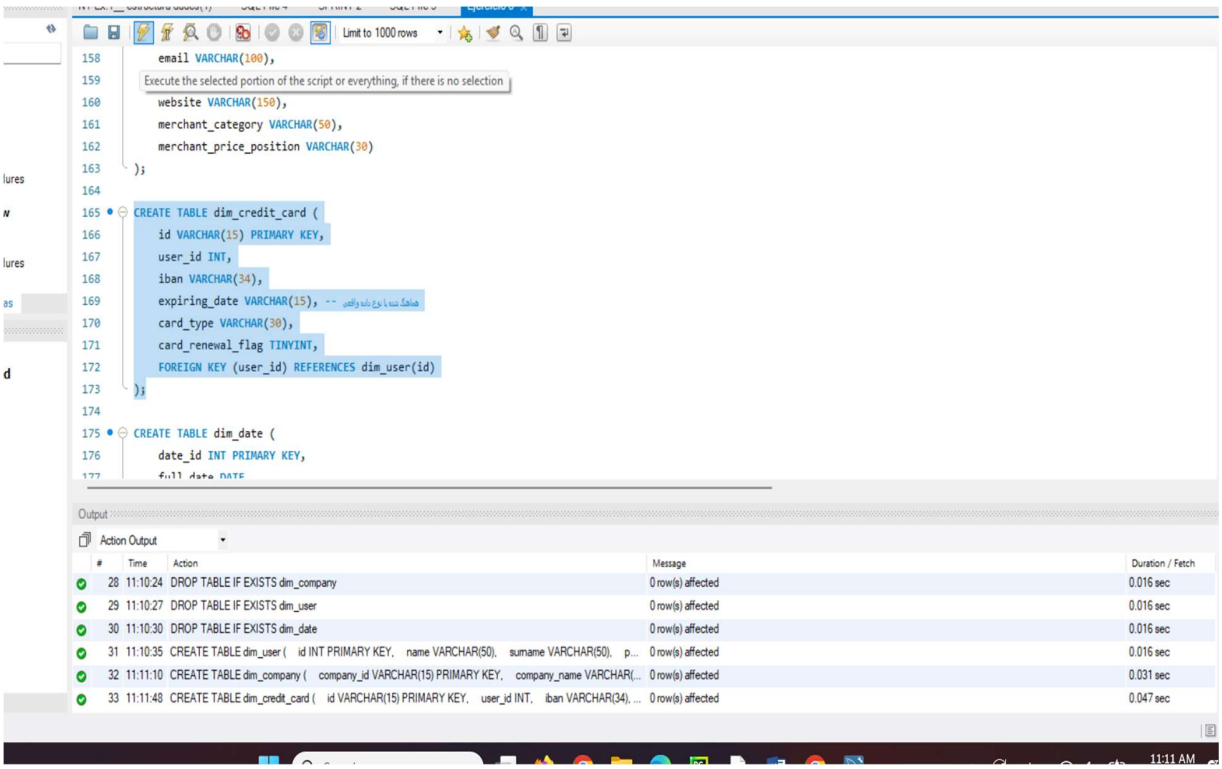
#	Time	Action	Message	Duration / Fetch
16	11:06:15	CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1...	Error Code: 1050. Table 'stg_transactions' already exists	0.000 sec
17	11:06:20	DROP TABLE IF EXISTS stg_transactions	0 row(s) affected	0.016 sec
18	11:06:27	CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1...	0 row(s) affected	0.016 sec
19	11:07:13	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv' INT...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0	0.032 sec
20	11:08:08	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv' IN...	3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0	0.093 sec
21	11:08:31	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv' INTO T...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.032 sec

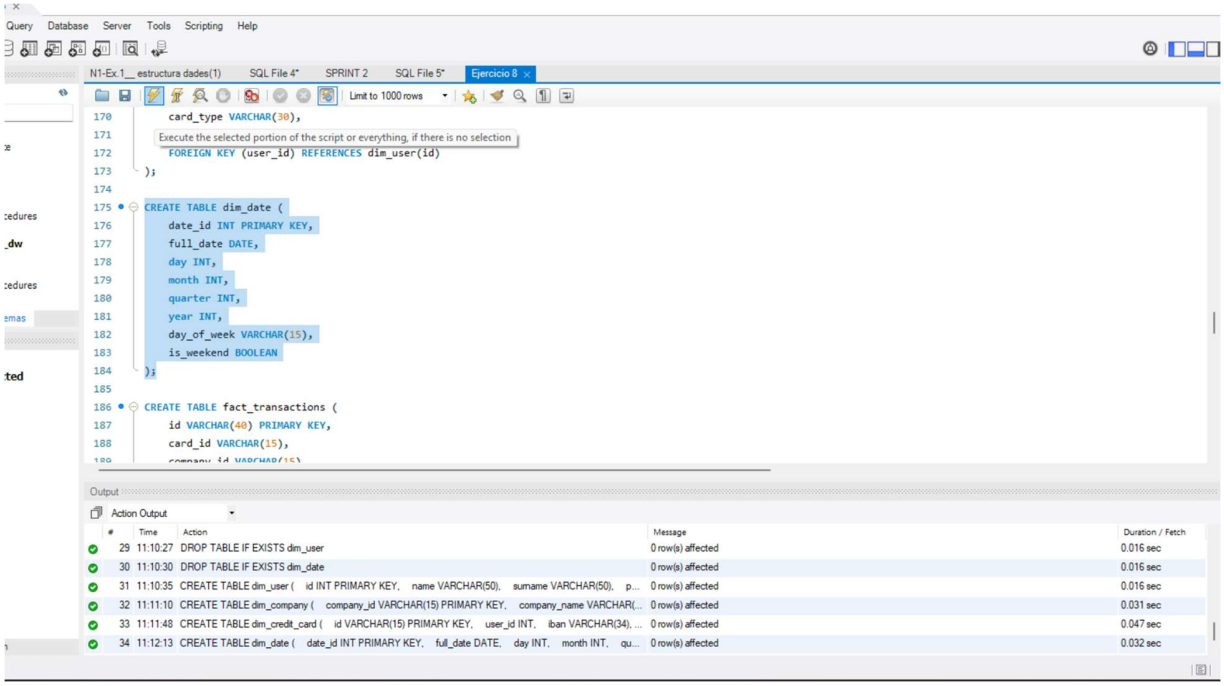

N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
Execute the selected portion of the script or everything, if there is no selection
104 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv'
105 INTO TABLE
106 FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
107 LINES TERMINATED BY '\n'
108 IGNORE 1 ROWS;
111 DROP TABLE IF EXISTS stg_credit_cards;
113 CREATE TABLE stg_credit_cards (
114 id VARCHAR(255),
115 user_id VARCHAR(255),
116 iban VARCHAR(255),
117 expiring_date VARCHAR(255),
118 card_type VARCHAR(255),
119 card_renewal_flag VARCHAR(255)
120);
122 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__credit_cards.csv'
Output
Action Output
Time Action Message Duration / Fetch
18 11:06:27 CREATE TABLE stg_transactions (id VARCHAR(40), card_id VARCHAR(15), business_id VARCHAR(1... 0 row(s) affected 0.016 sec
19 11:07:13 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv' INT... 1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0 0.032 sec
20 11:08:08 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv' IN... 3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0 0.093 sec
21 11:08:31 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv' INTO T... 100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0 0.032 sec
22 11:08:58 DROP TABLE IF EXISTS stg_credit_cards 0 row(s) affected 0.000 sec
23 11:09:02 CREATE TABLE stg_credit_cards (id VARCHAR(255), user_id VARCHAR(255), iban VARCHAR(255), ... 0 row(s) affected 0.031 sec

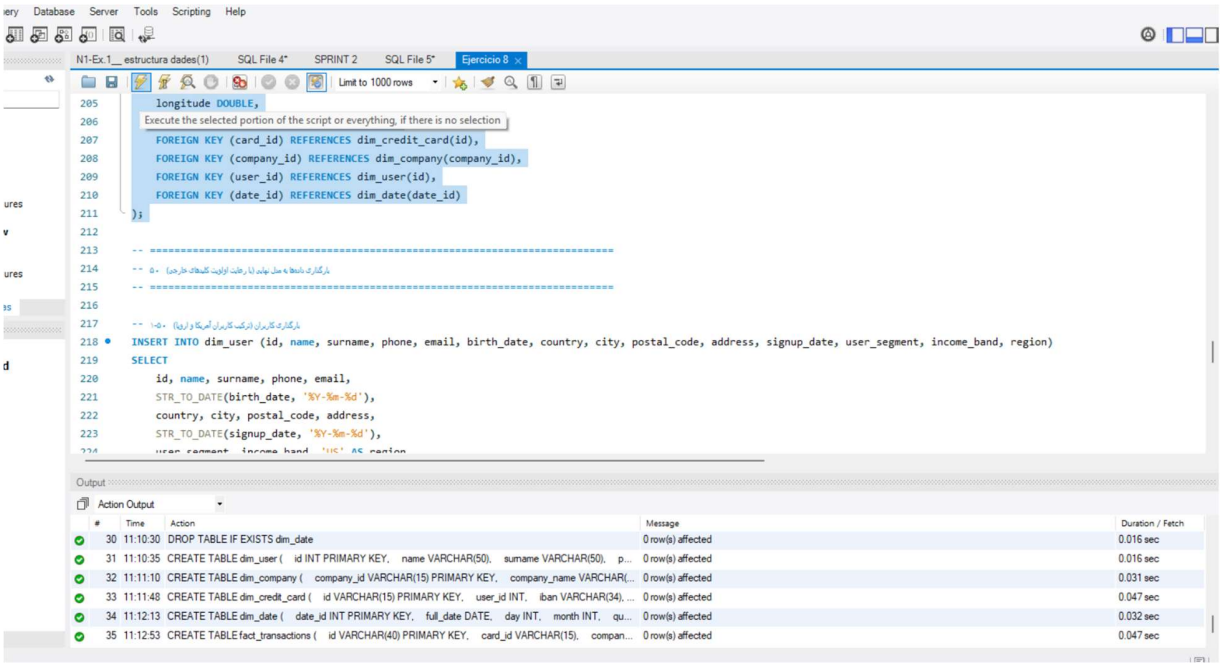

N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
Execute the selected portion of the script or everything, if there is no selection
116 iban VARCHAR(255),
118 card_type VARCHAR(255),
119 card_renewal_flag VARCHAR(255)
120);
122 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__credit_cards.csv'
123 INTO TABLE stg_credit_cards
124 FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
125 LINES TERMINATED BY '\r\n'
126 IGNORE 1 ROWS;
128 -- ==
129 -- ۴. ساخت جدول‌های مدل ابعادی (Star Schema)
130 -- ==
131 DROP TABLE IF EXISTS fact_transactions;
132 DROP TABLE IF EXISTS dim_credit_card;
133 DROP TABLE IF EXISTS dim_company;
134 DROP TABLE IF EXISTS dim_user;
Output
Action Output
Time Action Message Duration / Fetch
19 11:07:13 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv' INT... 1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0 0.032 sec
20 11:08:08 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv' IN... 3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0 0.093 sec
21 11:08:31 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv' INTO T... 100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0 0.032 sec
22 11:08:58 DROP TABLE IF EXISTS stg_credit_cards 0 row(s) affected 0.000 sec
23 11:09:02 CREATE TABLE stg_credit_cards (id VARCHAR(255), user_id VARCHAR(255), iban VARCHAR(255), ... 0 row(s) affected 0.031 sec
24 11:09:33 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__credit_cards.csv' INTO ... 0 row(s) affected Records: 0 Deleted: 0 Skipped: 0 Warnings: 0 0.015 sec

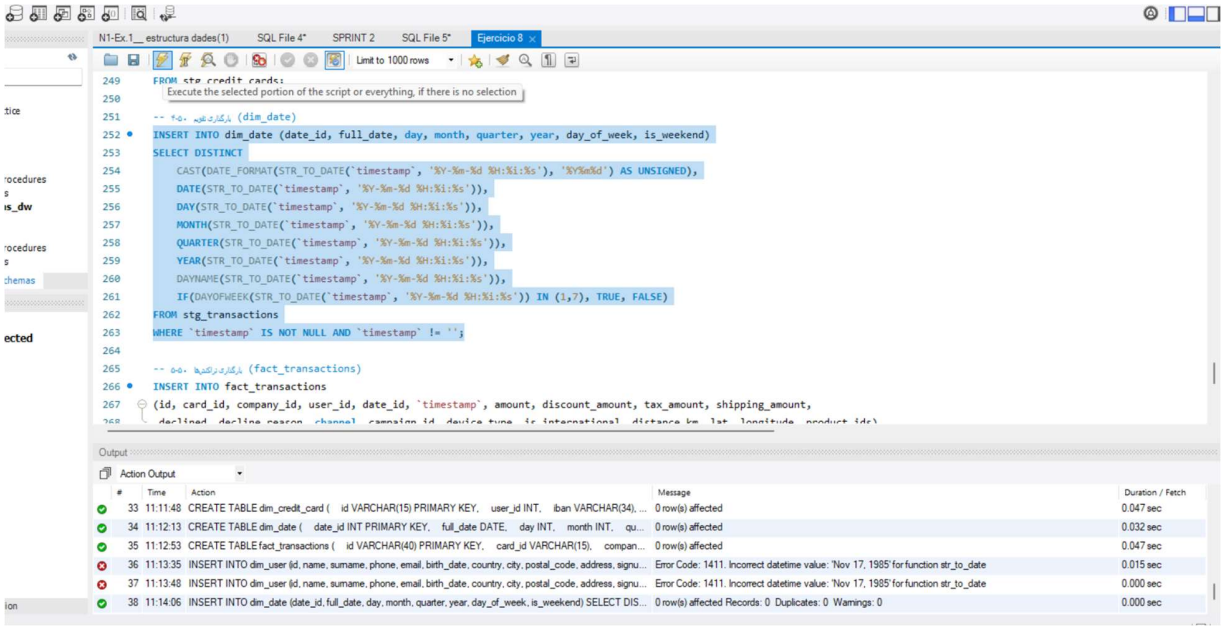

N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
DROP TABLE IF EXISTS dim_user;
DROP TABLE IF EXISTS dim_date;
CREATE TABLE dim_user (
id INT PRIMARY KEY,
name VARCHAR(50),
surname VARCHAR(50),
phone VARCHAR(30),
email VARCHAR(100),
birth_date DATE,
country VARCHAR(50),
city VARCHAR(50),
postal_code VARCHAR(20),
address VARCHAR(150),
signup_date DATE,
user_segment VARCHAR(50),
income_band VARCHAR(20),
region VARCHAR(20) -- نام ستون دقیقا مطابق با کد ثبت شده
);
Output
Action Output
Time Action Message Duration / Fetch
26 11:10:17 DROP TABLE IF EXISTS fact_transactions 0 row(s) affected 0.047 sec
27 11:10:20 DROP TABLE IF EXISTS dim_credit_card 0 row(s) affected 0.016 sec
28 11:10:24 DROP TABLE IF EXISTS dim_company 0 row(s) affected 0.016 sec
29 11:10:27 DROP TABLE IF EXISTS dim_user 0 row(s) affected 0.016 sec
30 11:10:30 DROP TABLE IF EXISTS dim_date 0 row(s) affected 0.016 sec
31 11:10:35 CREATE TABLE dim_user (id INT PRIMARY KEY, name VARCHAR(50), surname VARCHAR(50), p... 0 row(s) affected 0.016 sec

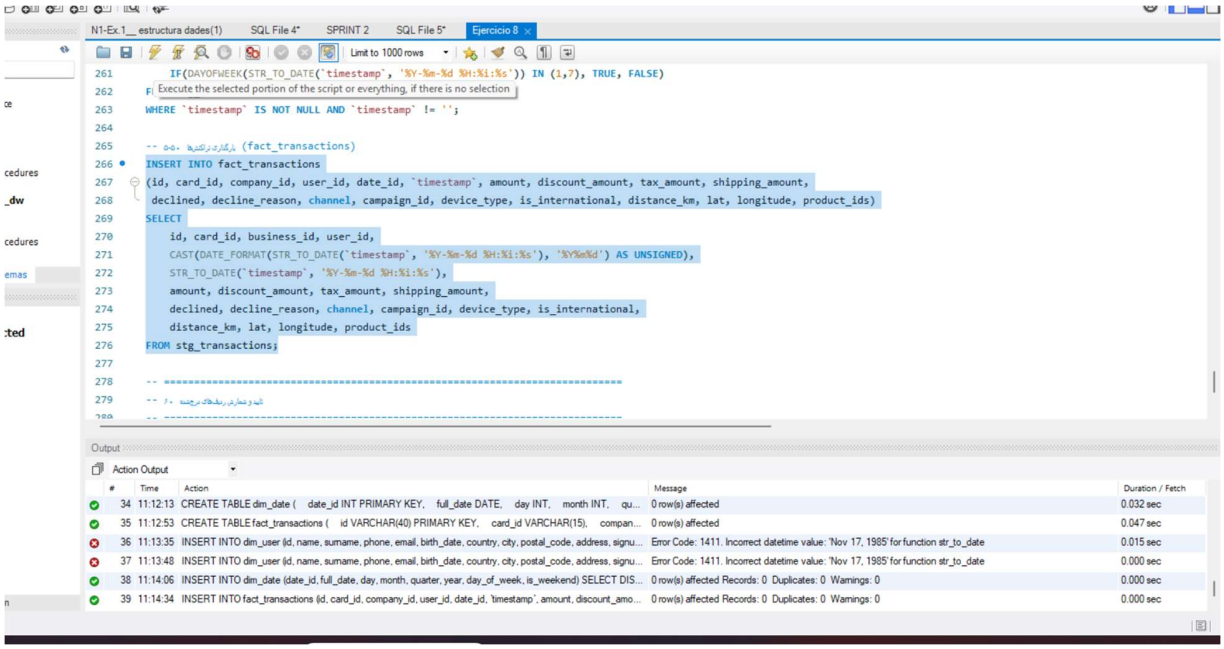

N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
Execute the selected portion of the script or everything, if there is no selection
CREATE TABLE dim_company (
company_id VARCHAR(15) PRIMARY KEY,
company_name VARCHAR(100),
phone VARCHAR(30),
email VARCHAR(100),
country VARCHAR(50),
website VARCHAR(150),
merchant_category VARCHAR(50),
merchant_price_position VARCHAR(30)
);
CREATE TABLE dim_credit_card (
id VARCHAR(15) PRIMARY KEY,
user_id INT,
iban VARCHAR(34),
expiring_date VARCHAR(15), -- هماهنگ شده با نوع داده واقعی
card_type VARCHAR(30),
Output
Action Output
Time Action Message Duration / Fetch
27 11:10:20 DROP TABLE IF EXISTS dim_credit_card 0 row(s) affected 0.016 sec
28 11:10:24 DROP TABLE IF EXISTS dim_company 0 row(s) affected 0.016 sec
29 11:10:27 DROP TABLE IF EXISTS dim_user 0 row(s) affected 0.016 sec
30 11:10:30 DROP TABLE IF EXISTS dim_date 0 row(s) affected 0.016 sec
31 11:10:35 CREATE TABLE dim_user (id INT PRIMARY KEY, name VARCHAR(50), surname VARCHAR(50), p... 0 row(s) affected 0.016 sec
32 11:11:10 CREATE TABLE dim_company (company_id VARCHAR(15) PRIMARY KEY, company_name VARCHAR(... 0 row(s) affected 0.031 sec

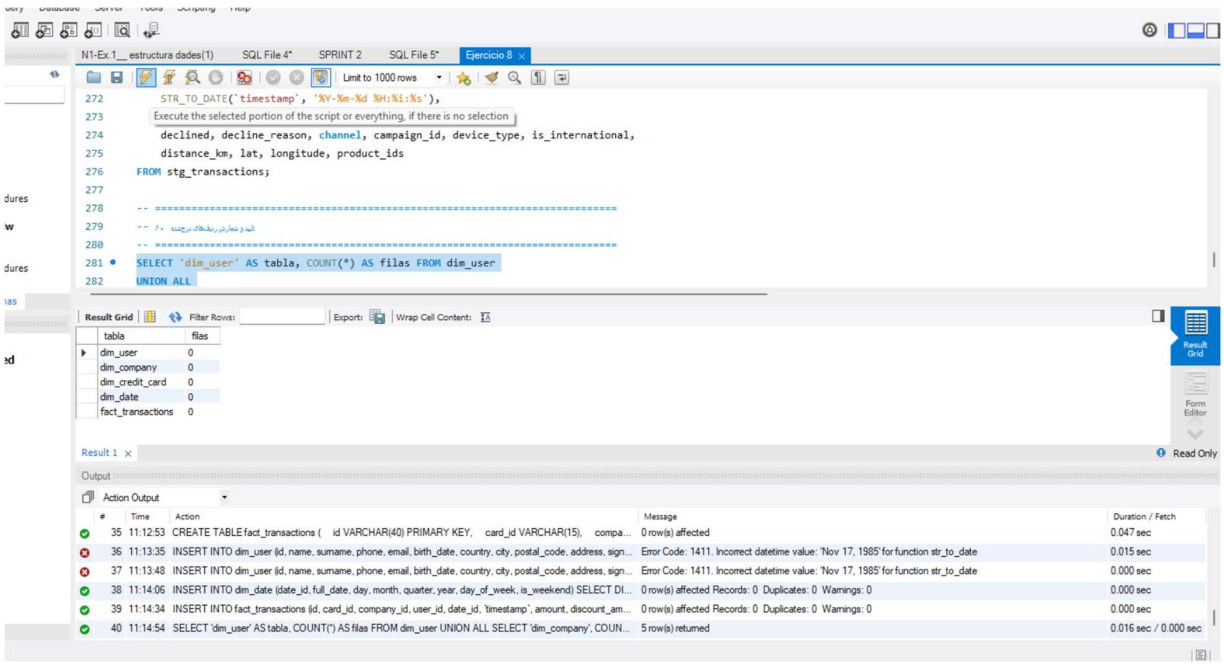

Limit to 1000 rows
email VARCHAR(100),
Execute the selected portion of the script or everything, if there is no selection
website VARCHAR(150),
merchant_category VARCHAR(50),
merchant_price_position VARCHAR(30)
);
CREATE TABLE dim_credit_card (
id VARCHAR(15) PRIMARY KEY,
user_id INT,
iban VARCHAR(34),
expiring_date VARCHAR(15), -- هماهنگ شده با نوع داده واقعی
card_type VARCHAR(30),
card_renewal_flag TINYINT,
FOREIGN KEY (user_id) REFERENCES dim_user(id)
);
CREATE TABLE dim_date (
date_id INT PRIMARY KEY,
Output
Action Output
Time Action Message Duration / Fetch
28 11:10:24 DROP TABLE IF EXISTS dim_company 0 row(s) affected 0.016 sec
29 11:10:27 DROP TABLE IF EXISTS dim_user 0 row(s) affected 0.016 sec
30 11:10:30 DROP TABLE IF EXISTS dim_date 0 row(s) affected 0.016 sec
31 11:10:35 CREATE TABLE dim_user (id INT PRIMARY KEY, name VARCHAR(50), surname VARCHAR(50), p... 0 row(s) affected 0.016 sec
32 11:11:10 CREATE TABLE dim_company (company_id VARCHAR(15) PRIMARY KEY, company_name VARCHAR(... 0 row(s) affected 0.031 sec
33 11:11:48 CREATE TABLE dim_credit_card (id VARCHAR(15) PRIMARY KEY, user_id INT, iban VARCHAR(34), ... 0 row(s) affected 0.047 sec


Query Database Server Tools Scripting Help
N1-Ex.1__ estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
card_type VARCHAR(30),
Execute the selected portion of the script or everything, if there is no selection
FOREIGN KEY (user_id) REFERENCES dim_user(id)
);
CREATE TABLE dim_date (
date_id INT PRIMARY KEY,
full_date DATE,
day INT,
month INT,
quarter INT,
year INT,
day_of_week VARCHAR(15),
is_weekend BOOLEAN
);
CREATE TABLE fact_transactions (
id VARCHAR(40) PRIMARY KEY,
card_id VARCHAR(15),
Output
Action Output
Time Action Message Duration / Fetch
29 11:10:27 DROP TABLE IF EXISTS dim_user 0 row(s) affected 0.016 sec
30 11:10:30 DROP TABLE IF EXISTS dim_date 0 row(s) affected 0.016 sec
31 11:10:35 CREATE TABLE dim_user (id INT PRIMARY KEY, name VARCHAR(50), surname VARCHAR(50), p... 0 row(s) affected 0.016 sec
32 11:11:10 CREATE TABLE dim_company (company_id VARCHAR(15) PRIMARY KEY, company_name VARCHAR(... 0 row(s) affected 0.031 sec
33 11:11:48 CREATE TABLE dim_credit_card (id VARCHAR(15) PRIMARY KEY, user_id INT, iban VARCHAR(34), ... 0 row(s) affected 0.047 sec
34 11:12:13 CREATE TABLE dim_date (date_id INT PRIMARY KEY, full_date DATE, day INT, month INT, qu... 0 row(s) affected 0.032 sec


Database Server Tools Scripting Help
N1-Ex.1__ estructura dades(1) SQL File 4* SPRINT 2 SQL File 5* Ejercicio 8
Limit to 1000 rows
longitude DOUBLE,
Execute the selected portion of the script or everything, if there is no selection
FOREIGN KEY (card_id) REFERENCES dim_credit_card(id),
FOREIGN KEY (company_id) REFERENCES dim_company(company_id),
FOREIGN KEY (user_id) REFERENCES dim_user(id),
FOREIGN KEY (date_id) REFERENCES dim_date(date_id)
);
-- ==========
-- ۵۰ بارگذاری داده‌ها به جدل نهایی (با رعایت اولویت کلیدهای خارجی)
-- ==========
-- ۵۰.۱ بارگذاری کاربران (ترکیب کاربران آمریکا و اروپا)
INSERT INTO dim_user (id, name, surname, phone, email, birth_date, country, city, postal_code, address, signup_date, user_segment, income_band, region)
SELECT
id, name, surname, phone, email,
STR_TO_DATE(birth_date, '%Y-%m-%d'),
country, city, postal_code, address,
STR_TO_DATE(signup_date, '%Y-%m-%d'),
Output
Action Output
Time Action Message Duration / Fetch
30 11:10:30 DROP TABLE IF EXISTS dim_date 0 row(s) affected 0.016 sec
31 11:10:35 CREATE TABLE dim_user (id INT PRIMARY KEY, name VARCHAR(50), surname VARCHAR(50), p... 0 row(s) affected 0.016 sec
32 11:11:10 CREATE TABLE dim_company (company_id VARCHAR(15) PRIMARY KEY, company_name VARCHAR(... 0 row(s) affected 0.031 sec
33 11:11:48 CREATE TABLE dim_credit_card (id VARCHAR(15) PRIMARY KEY, user_id INT, iban VARCHAR(34), ... 0 row(s) affected 0.047 sec
34 11:12:13 CREATE TABLE dim_date (date_id INT PRIMARY KEY, full_date DATE, day INT, month INT, qu... 0 row(s) affected 0.032 sec
35 11:12:53 CREATE TABLE fact_transactions (id VARCHAR(40) PRIMARY KEY, card_id VARCHAR(15), compan... 0 row(s) affected 0.047 sec


N1-Ex.1__ estructura dades(1) SQL File 4* SPRINT 2 SQL File 5* Ejercicio 8
Limit to 1000 rows
FROM stg_credit_cards;
Execute the selected portion of the script or everything, if there is no selection
-- ۴۵۰ بارگذاری تقویم (dim_date)
INSERT INTO dim_date (date_id, full_date, day, month, quarter, year, day_of_week, is_weekend)
SELECT DISTINCT
CAST(DATE_FORMAT(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s'), '%Y%m%d') AS UNSIGNED),
DATE(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s')),
DAY(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s')),
MONTH(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s')),
QUARTER(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s')),
YEAR(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s')),
DAYNAME(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s')),
IF(DAYOFWEEK(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s')) IN (1,7), TRUE, FALSE)
FROM stg_transactions
WHERE `timestamp` IS NOT NULL AND `timestamp` != '';
-- ۵۰۰ بارگذاری تراکنش‌ها (fact_transactions)
INSERT INTO fact_transactions
(id, card_id, company_id, user_id, date_id, `timestamp`, amount, discount_amount, tax_amount, shipping_amount,
Output
Action Output
Time Action Message Duration / Fetch
33 11:11:48 CREATE TABLE dim_credit_card (id VARCHAR(15) PRIMARY KEY, user_id INT, iban VARCHAR(34), ... 0 row(s) affected 0.047 sec
34 11:12:13 CREATE TABLE dim_date (date_id INT PRIMARY KEY, full_date DATE, day INT, month INT, qu... 0 row(s) affected 0.032 sec
35 11:12:53 CREATE TABLE fact_transactions (id VARCHAR(40) PRIMARY KEY, card_id VARCHAR(15), compan... 0 row(s) affected 0.047 sec
36 11:13:35 INSERT INTO dim_user (id, name, surname, phone, email, birth_date, country, city, postal_code, address, signu... Error Code: 1411. Incorrect datetime value: 'Nov 17, 1985' for function str_to_date 0.015 sec
37 11:13:48 INSERT INTO dim_user (id, name, surname, phone, email, birth_date, country, city, postal_code, address, signu... Error Code: 1411. Incorrect datetime value: 'Nov 17, 1985' for function str_to_date 0.000 sec
38 11:14:06 INSERT INTO dim_date (date_id, full_date, day, month, quarter, year, day_of_week, is_weekend) SELECT DIS... 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.000 sec


N1-Ex.1__estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
Execute the selected portion of the script or everything, if there is no selection
IF(DAYOFWEEK(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s')) IN (1,7), TRUE, FALSE)
WHERE `timestamp` IS NOT NULL AND `timestamp` != '';
-- ۵۵۰ بارگذاری رکوردها (fact_transactions)
INSERT INTO fact_transactions
(id, card_id, company_id, user_id, date_id, `timestamp`, amount, discount_amount, tax_amount, shipping_amount,
declined, decline_reason, channel, campaign_id, device_type, is_international, distance_km, lat, longitude, product_ids)
SELECT
id, card_id, business_id, user_id,
CAST(DATE_FORMAT(STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s'), '%Y%m%d') AS UNSIGNED),
STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s'),
amount, discount_amount, tax_amount, shipping_amount,
declined, decline_reason, channel, campaign_id, device_type, is_international,
distance_km, lat, longitude, product_ids
FROM stg_transactions;
-- ۶۰ تایید و شمارش ردیف‌های درج‌شده
Output
Action Output
Time Action Message Duration / Fetch
34 11:12:13 CREATE TABLE dim_date (date_id INT PRIMARY KEY, full_date DATE, day INT, month INT, qu... 0 row(s) affected 0.032 sec
35 11:12:53 CREATE TABLE fact_transactions (id VARCHAR(40) PRIMARY KEY, card_id VARCHAR(15), compan... 0 row(s) affected 0.047 sec
36 11:13:35 INSERT INTO dim_user (id, name, surname, phone, email, birth_date, country, city, postal_code, address, signu... Error Code: 1411. Incorrect datetime value: 'Nov 17, 1985' for function str_to_date 0.015 sec
37 11:13:48 INSERT INTO dim_user (id, name, surname, phone, email, birth_date, country, city, postal_code, address, signu... Error Code: 1411. Incorrect datetime value: 'Nov 17, 1985' for function str_to_date 0.000 sec
38 11:14:06 INSERT INTO dim_date (date_id, full_date, day, month, quarter, year, day_of_week, is_weekend) SELECT DIS... 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.000 sec
39 11:14:34 INSERT INTO fact_transactions (id, card_id, company_id, user_id, date_id, `timestamp`, amount, discount_amo... 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.000 sec


N1-Ex.1__estructura dades(1)
SQL File 4*
SPRINT 2
SQL File 5*
Ejercicio 8
Limit to 1000 rows
Execute the selected portion of the script or everything, if there is no selection
STR_TO_DATE(`timestamp`, '%Y-%m-%d %H:%i:%s'),
declined, decline_reason, channel, campaign_id, device_type, is_international,
distance_km, lat, longitude, product_ids
FROM stg_transactions;
-- ۶۰ تایید و شمارش ردیف‌های درج‌شده
SELECT 'dim_user' AS tabla, COUNT(*) AS filas FROM dim_user
UNION ALL
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
tabla filas
dim_user 0
dim_company 0
dim_credit_card 0
dim_date 0
fact_transactions 0
Result 1
Read Only
Output
Action Output
Time Action Message Duration / Fetch
35 11:12:53 CREATE TABLE fact_transactions (id VARCHAR(40) PRIMARY KEY, card_id VARCHAR(15), compa... 0 row(s) affected 0.047 sec
36 11:13:35 INSERT INTO dim_user (id, name, surname, phone, email, birth_date, country, city, postal_code, address, sign... Error Code: 1411. Incorrect datetime value: 'Nov 17, 1985' for function str_to_date 0.015 sec
37 11:13:48 INSERT INTO dim_user (id, name, surname, phone, email, birth_date, country, city, postal_code, address, sign... Error Code: 1411. Incorrect datetime value: 'Nov 17, 1985' for function str_to_date 0.000 sec
38 11:14:06 INSERT INTO dim_date (date_id, full_date, day, month, quarter, year, day_of_week, is_weekend) SELECT DI... 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.000 sec
39 11:14:34 INSERT INTO fact_transactions (id, card_id, company_id, user_id, date_id, `timestamp`, amount, discount_am... 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.000 sec
40 11:14:54 SELECT 'dim_user' AS tabla, COUNT(*) AS filas FROM dim_user UNION ALL SELECT 'dim_company', COUN... 5 row(s) returned 0.016 sec / 0.000 sec